

Three-Year Financial Model

Every figure derived from the drivers below — nothing is asserted

1 Drivers

This is the only tab with typed-in numbers. Everything on pages 2–4 is a formula pointing back here, so changing one assumption moves the whole model. The three revenue targets carried over from the Round 1 narrative — \$1.0M, \$8.2M, \$36.1M — are not typed anywhere; they fall out of these inputs.

DRIVER	2026	2027	2028	BASIS / SOURCE
VOLUME				
Provider groups — ending	110	560	2,240	
Provider groups — average active	55	330	1,400	
Physicians per group	8	8	8	
Prior auths per physician per week	30	30	30	
Share of PA volume routed through Pavo	25%	40%	50%	
Payer contracts — average active	11	84	160	
PRICE				
Fee per authorization	\$3.00	\$3.00	\$3.00	
Provider subscription	\$400	\$400	\$400	
Payer compliance API	\$20,000	\$20,000	\$20,000	
ACQUISITION				
New provider groups acquired	110	470	1,750	
New payer contracts acquired	18	122	80	
Provider CAC	\$9,500	\$7,200	\$6,000	
Payer CAC	\$34,000	\$27,000	\$22,000	
Provider annual logo churn	8%	6%	5%	
Payer annual logo churn	2%	2%	2%	
COST OF REVENUE				
Cloud infrastructure	\$54,000	\$198,000	\$900,000	
Model API (Insure + appeals only)	\$11,000	\$62,000	\$280,000	
Clinical rule library — FTE	1.5	3.5	8.0	
Clinical rule library — loaded cost/FTE	\$126,000	\$130,000	\$135,000	
Implementation & onboarding — FTE	2.0	4.0	12.0	
Implementation — loaded cost/FTE	\$112,000	\$119,000	\$119,000	
Customer support — FTE	1.0	2.0	8.0	
Support — loaded cost/FTE	\$92,000	\$95,000	\$98,000	
Security & compliance	\$118,000	\$112,000	\$290,000	
OPERATING EXPENSE				
Engineers	4	12	26	
Loaded cost per engineer	\$170,000	\$175,000	\$180,000	
General & administrative	\$340,000	\$1,150,000	\$3,200,000	

Reading this model: every figure on tabs 2–7 is an Excel formula pointing back at this tab. The three revenue targets carried over from the Round 1 narrative (\$1.0M / \$8.2M / \$36.1M) are not typed anywhere — they fall out of the drivers above. Tab 7 shows what happens when the four most fragile of those drivers are wrong.

2 Revenue build

LINE	2026	2027	2028	NOTE
AUTHORIZATION VOLUME				
Prior auths per group per year	12,480	12,480	12,480	
Authorizations routed through Pavo, per group	3,120	4,992	6,240	
Total authorizations processed	171,600	1,647,360	8,736,000	
REVENUE BY STREAM				
Per-authorization fees	\$514,800	\$4,942,080	\$26,208,000	
Provider subscriptions	\$264,000	\$1,584,000	\$6,720,000	
Payer compliance API	\$220,000	\$1,680,000	\$3,200,000	
Total revenue	\$998,800	\$8,206,080	\$36,128,000	
Year-on-year growth		722%	340%	
MIX				
Authorization fees, share of revenue	51.5%	60.2%	72.5%	
Subscriptions, share of revenue	26.4%	19.3%	18.6%	
Compliance API, share of revenue	22.0%	20.5%	8.9%	

The authorization fee is 73% of 2028 revenue, which is why the model's single greatest exposure is provider group count rather than price. Tab 7 quantifies that.

3 Cost of revenue and gross margin

LINE	2026	2027	2028	DRIVER
Cloud infrastructure	\$54,000	\$198,000	\$900,000	Compute, Postgres, storage, egress, 7-year retention
Model API — Insure parsing, appeal drafting	\$11,000	\$62,000	\$280,000	Not on the authorization path
Clinical rule library	\$189,000	\$455,000	\$1,080,000	1.5 → 3.5 → 8.0 FTE informaticists
Implementation & onboarding	\$224,000	\$476,000	\$1,428,000	2.0 → 4.0 → 12.0 FTE
Customer support	\$92,000	\$190,000	\$784,000	1.0 → 2.0 → 8.0 FTE
Security & compliance	\$118,000	\$112,000	\$290,000	SOC 2 Type II, penetration testing, BAA management
Total cost of revenue	\$688,000	\$1,493,000	\$4,762,000	
Revenue	\$998,800	\$8,206,080	\$36,128,000	
Gross profit	\$310,800	\$6,713,080	\$31,366,000	
Gross margin	31%	82%	87%	

COST SHARE

People (rule library + implementation + support)	73.4%	75.1%	69.1%
Infrastructure + model API	9.4%	17.4%	24.8%

Year 1 gross margin of 31% is deliberate and corrects the Round 1 narrative, which implied break-even. Early pilots carry a disproportionate implementation load and are delivered near cost. Margin reaches 87% by 2028 because authorization volume scales without adding people — the marginal cost of one authorization is about \$0.00057 (tab 6).

4 Profit and loss

LINE	2026	2027	2028
Revenue	\$998,800	\$8,206,080	\$36,128,000
Cost of revenue	(\$688,000)	(\$1,493,000)	(\$4,762,000)
Gross profit	\$310,800	\$6,713,080	\$31,366,000
Gross margin	31%	82%	87%
OPERATING EXPENSE			
Research & development	(\$680,000)	(\$2,100,000)	(\$4,680,000)
Sales & marketing	(\$1,657,000)	(\$6,678,000)	(\$12,260,000)
New logos acquired	128	592	1,830
General & administrative	(\$340,000)	(\$1,150,000)	(\$3,200,000)
Total operating expense	(\$2,677,000)	(\$9,928,000)	(\$20,140,000)
EBITDA	(\$2,366,200)	(\$3,214,920)	\$11,226,000
EBITDA margin	-237%	-39%	31%
Cumulative EBITDA	(\$2,366,200)	(\$5,581,120)	\$5,644,880

Peak cumulative burn is \$5.58M. An \$8.0M raise covers it with roughly twelve months of buffer past EBITDA breakeven. Sales & marketing is the largest operating line in all three years, and 73% of it is provider acquisition — which the payer channel is what makes cheap.

5 Unit economics — CAC, LTV, payback

METRIC	PROVIDER GROUP	PAYER	BLENDED
Annual revenue per account	\$23,520	\$20,000	\$23,159
Gross margin applied	87%	87%	87%
Gross profit per account	\$20,420	\$17,364	\$20,106
Annual logo churn	5.0%	2.0%	4.7%
Customer acquisition cost	\$6,000	\$22,000	\$6,699
LIFETIME VALUE			
Year 1 gross profit	\$20,420	\$17,364	\$20,106
Year 2 gross profit	\$19,399	\$17,017	\$19,163
Year 3 gross profit	\$18,429	\$16,676	\$18,264
Year 4 gross profit	\$17,507	\$16,343	\$17,407
Year 5 gross profit	\$16,632	\$16,016	\$16,590
LTV — 3-year	\$58,248	\$51,057	\$57,533
LTV — 5-year	\$92,387	\$83,415	\$91,530
LTV : CAC (3-year)	9.7×	2.3×	8.6×
CAC payback	3.5 mo	15.2 mo	4.0 mo

Read the payer column correctly. At 2.3× LTV:CAC and a 15-month payback, payer contracts do not pay for themselves as a standalone product, and we are not pretending otherwise. The payer contract is a distribution channel: signing one payer makes Pavo reachable by every provider group already submitting authorizations to it, which is why provider CAC falls 37% across the plan. Judged alone the payer line looks weak; judged as the thing that makes the provider line work, it is the highest-leverage spend in the model.

6 Server and API cost — measured, not assumed

Timings taken against the committed code on 13 September 2026. The point is not that infrastructure is cheap — it is that infrastructure is not what this business spends money on. People encoding payer criteria are.

COMPONENT	MEASURED	UNIT COST	COST PER AUTH	HOW IT WAS MEASURED
MEASURED PERFORMANCE				
Authorization decision — p50 latency	0.0886 s	20 consecutive POST /api/auth/request calls		
Authorization decision — p95 latency	0.1133 s			
Agent-to-agent resolution	0.0447 s	average_resolution_seconds from /api/system/stats		
ZK proof generation	0.406 s	Groth16, 5 runs		
ZK proof verification	0.409 s			
Database rows written per authorization	8	1 auth_request + 2 aria_messages + 5 audit_log		
Payload written per authorization (kB)	6			

MARGINAL COST OF ONE AUTHORIZATION

Component	Quantity	Unit cost	Cost per auth	Note
Compute — vCPU seconds	0.0886	\$0.000013	\$0.000001	\$0.045/vCPU-hour on a managed container platform
Database write + 7-year retention	0.000006	\$10.50	\$0.000063	6 kB held 84 months at \$0.125/GB/month
ZK proof (20% of authorizations)	0.0812	\$0.000013	\$0.000001	Short-lived Node process; only when the payer requires it
Model API tokens	0	\$0.000000	\$0.000000	No inference on the authorization path — by design
Load balancing, logging, observability overhead	1	\$0.000500	\$0.000500	Allocated; dominates the measured components
Marginal cost per authorization			\$0.000565	
Revenue per authorization			\$3.00	
Marginal gross margin per authorization			99.98%	

ANNUAL INFRASTRUCTURE BUDGET vs. MARGINAL COST

Authorizations processed	171,600	1,647,360	8,736,000
Marginal cost at scale	\$97	\$931	\$4,937
Budgeted cloud infrastructure	\$54,000	\$198,000	\$900,000
Headroom (budget less marginal)	\$53,903	\$197,069	\$895,063

The budget deliberately exceeds the marginal cost by a wide margin: it carries baseline capacity, staging and disaster-recovery environments, observability, and the reserved headroom an availability commitment to a payer requires. The point of this tab is not that infrastructure is cheap — it is that infrastructure is not what this business spends money on. People encoding payer criteria are (tab 3).

7 Sensitivity

SCENARIO	PROVIDER GROUPS	ROUTED SHARE	FEE PER AUTH	PAYERS	2028 REVENUE	VS. BASE
Base case	1,400	50%	\$3.00	160	\$36,128,000	0%
CMS-0057-F slips a year → payer count halves	1,400	50%	\$3.00	80	\$34,528,000	-4%
Routed share is 35%, not 50%	1,400	35%	\$3.00	160	\$28,265,600	-22%
Provider groups reach 900 average, not 1,400	900	50%	\$3.00	160	\$24,368,000	-33%
Authorization fee compresses to \$2.00	1,400	50%	\$2.00	160	\$27,392,000	-24%
All four together	900	35%	\$2.00	80	\$13,782,400	-62%

Provider group count is the model's single greatest exposure — halving it costs more than halving the payer count and compressing price combined, because authorization fees are 72% of 2028 revenue. It is also the driver most within our control, which is why the ask in the report weights payer business development so heavily: the payer channel is what makes provider acquisition affordable at that scale.